Findings Summary

A transparent safe-system speed-limit alignment method

Submitted by the University of New South Wales (UNSW) team

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Executive summary

This work answers the challenge question directly on **where the posted speed limits are misaligned with road function and vulnerable-road-user (VRU) exposure?** The focus is not on whether drivers are speeding, but on whether the posted limit itself is appropriate for the road. Pedestrians and cyclists are the least protected road users in a crash. At higher speeds, even a minor conflict can become fatal for them. So, whether vulnerable road users are present in the street context is key to judging whether a speed limit is safe.

For every road link, we classify the road setting, assign the Safe-System speed that fits that setting, and recommend a reduction wherever the posted limit is higher. Where street-level imagery is available, an AI model reads the scene to improve the classification. The result is a practical and transparent recommendation for each link that includes a proposed speed limit, the size of the speed misalignment, and a risk flag. It is designed to be understood by non-technical decision-makers and to scale across cities and countries.

The two study areas tell very different stories. In Thailand, around nine in ten links on the road network have posted speed limits above Safe-System levels and are therefore recommended for reduction, with a typical cut of about 20 km/h. In Maharashtra, only around 15% need to change, and usually by a small margin of around 5 km/h. Street imagery confirms that the highest-priority roads are genuine high-speed, high-conflict environments, such as fast highways that still have pedestrians, roadside activity and at-grade junctions, rather than noise in the input data.

The question and the Safe-System reference

Speed is one of the main factors that determines whether a crash is survivable. The Safe System approach starts from a simple principle. Speed limits should be set so that, if a crash does happen, it is more likely to be survivable for the most vulnerable road user present. For this work, we translated the published Safe-System guidance into the following reference values, based on internationally recognised references including WHO Speed Management[1], ITF Speed and Crash Risk[2], and the World Bank GRSF Guide for Safe Speeds[3]:

Road context	Appropriate maximum speed
Pedestrians and cyclists mixing with traffic	30 km/hr
Urban, likely VRU presence, at-grade intersections	50 km/hr
Rural undivided (head-on/run-off risk)	70 km/hr
Divided/limited-access motorway, no VRU exposure	100–110 km/hr

We also use a hard function gate for motorways and divided roads with no pedestrians and cyclists exposure. These roads are assigned the higher Safe-System value and are not flagged simply because vehicles are travelling fast. This is important because a high operating speed should not, on its own, trigger a lower limit on a road that is genuinely designed for high-speed movement.

Method (transparent, five steps)

The method is deliberately simple, scalable and auditable. For each road link, we do five things:

1. **Classify the road context** using road function and land use, then refine this with street imagery where it is available.
2. **Assign the appropriate Safe-System maximum** from the reference table above.
3. **Recommend a reduction only.** The recommended limit is the lower of the posted limit and the Safe-System maximum.
4. **Calculate the misalignment.** This is the difference between the current posted limit and the recommended limit.
5. **Flag the highest-risk links.** A link is marked High where the current limit is materially too high for a vulnerable road user context, and the observed 85th-percentile speed confirms that the risk is actually present on the road.

(Left) *A road the data labels RURAL. The photo shows pedestrians/cyclists mixing with traffic, so it is re-classified to 30 km/h and flagged.* ***(Right)*** *A road labelled as RURAL is actually a divided, limited-access road. It is classified in the 100 km/h band and cleared.*

[1] World Health Organisation. Speed management: a road safety manual for decision-makers and practitioners, 2nd edition.
[2] International Transport Forum. Speed and Crash Risk.
[3] World Bank Group. Detecting Urban Clues for Road Safety.

Imagery refinement: Fixing the data's known weaknesses

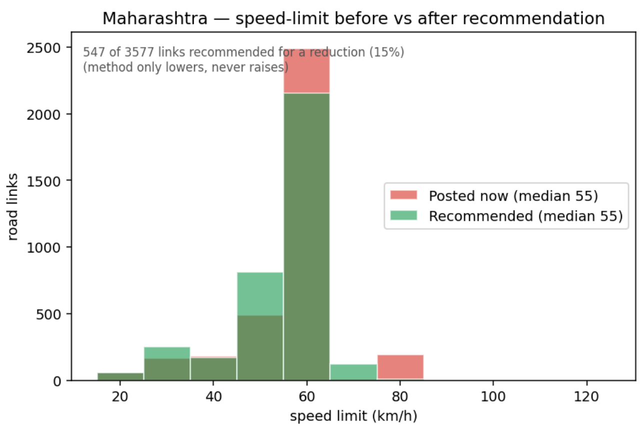
The challenge instructions made clear that the provided land use and speed limit fields are estimates, not ground truth. They need to be checked against what is actually visible on the road. Our method does this directly.

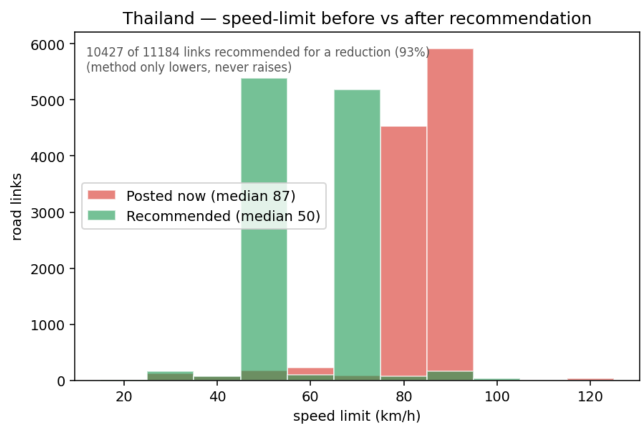
Where Mapillary imagery is available, a vision-language model (OpenAI Chat GPT 5.5) reads the street photo and returns the road context[4]. We keep this imagery-based classification next to the provided data, rather than overwriting the original information. This makes the process easier to audit. The supplied data, the imagery interpretation, and the final classification can all be compared link by link.

The imagery refinement helps in two directions. In one case, a link labelled "RURAL" in the data clearly shows pedestrians and cyclists mixing with traffic, so it is reclassified to the 30 km/h band and flagged. In another case, a "RURAL" link is actually a divided, limited-access road, so it is moved to the 100 km/h band and cleared.

Results

The two study areas show very different patterns, and the difference is mainly driven by the posted limits rather than the road network itself. In Thailand, most higher-order roads are posted at 80–90 km/h. Around 93% of links are above the Safe-System level and are recommended for reduction. This reflects a network where posted limits appear to be set mainly for traffic flow, even where the road context includes urban activity, intersections or likely pedestrians and cyclists exposure. Maharashtra is very different. Posted speed limits are mostly clustered around 55 km/h, and only around 15% of roads are recommended for reduction. In most cases, the existing limits are already close to, or below, the Safe-System ceilings.





*Posted and recommended speed limits, showing major reductions in Thailand **(right)** and limited change in Maharashtra **(left)**.*

Summary of posted versus recommended speed limits and priority outcomes in Thailand and Maharashtra.

	Thailand	Maharashtra
Number of road links (with a posted speed limit)	11,184	3,577
Median posted speed limit	87 km/h	55 km/h
Recommended for a reduction	95%	15%
Median recommended speed limit	50 km/h	55 km/h
High-priority road links	4,696 (42%)	130 (4%)
Confirmed (operating-speed + imagery)	3,171	125
Imagery-confirmed high-speed conflict	858	41
Needs expert review (no imagery to verify)	1,525	5

[4] Jongwiriyanurak et al. (2025). V-RoAst: Visual Road Assessment. Can VLM be a Road Safety Assessor Using the iRAP Standard? Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops.

Validation, robustness and honest limitations

The method only recommends reductions. The Safe-System value is treated as a ceiling, not a target.

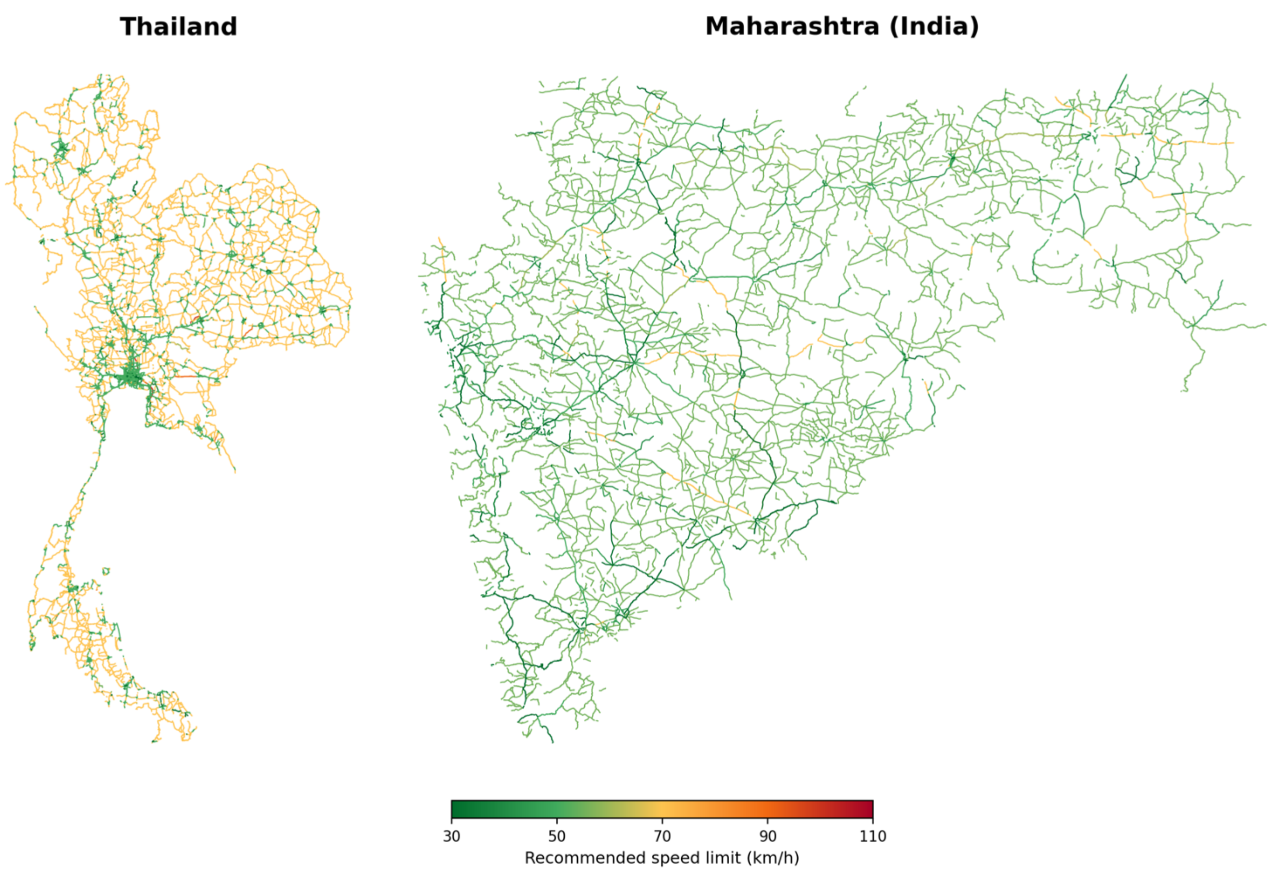
It is also self-auditing against operating speed. A finding is only treated as confident when the observed 85th-percentile speed is plausible for the assigned class. Where it is not, the link is flagged for verification rather than reported as high risk. This helps guard against land-use estimation errors.

Where Mapillary imagery is available, the AI-based context is used to refine the classification. Elsewhere, the method falls back to a transparent data prior based on road class and land use. Each link keeps this status visible, so the result remains auditable.

Coverage is still limited, especially in rural India, where Mapillary is sparse. That means the score there leans more heavily on the data prior. A denser imagery source would improve the refinement.

There is also no crash data available, so this is a Safe-System appropriateness screen rather than an empirical crash model. The output should be treated as a prioritised review list and confirmed on site.

Finally, the challenge data itself comes with caveats. Speed limits and land use are estimates, and speeds are sampled along long segments. A sample-size sensitivity check is recommended before acting on individual links.



*Recommended speed limits per road link, Thailand **(left)** and Maharashtra **(right)**. Colour shows the Safe-System-appropriate limit, from green 30 km/h to red 110 km/h). Maharashtra reads predominantly green because its posted limits are already low (around 50–55 km/h) and change little; Thailand is a green–amber mix as urban and intersection roads are brought down to 50 km/h and rural roads to 70 km/h, with limited-access motorways (red) retained at 100 km/h.*

Scalability and recommendation

The method is country-agnostic. It only needs functional class, operating speed, land use and street imagery, all of which are often globally available, and it is interpretable.

In Thailand, the method identifies a large and structured opportunity to bring higher-order speed limits in line with Safe-System levels, especially along high-risk vulnerable road user corridors. In Maharashtra, it confirms broad alignment, with only a small, targeted shortlist needing expert review.

The method provides an interactive map that lets an official compare posted and recommended limits, filter to high-risk roads, and open the street photo behind each recommendation to verify and make a data-driven decision.